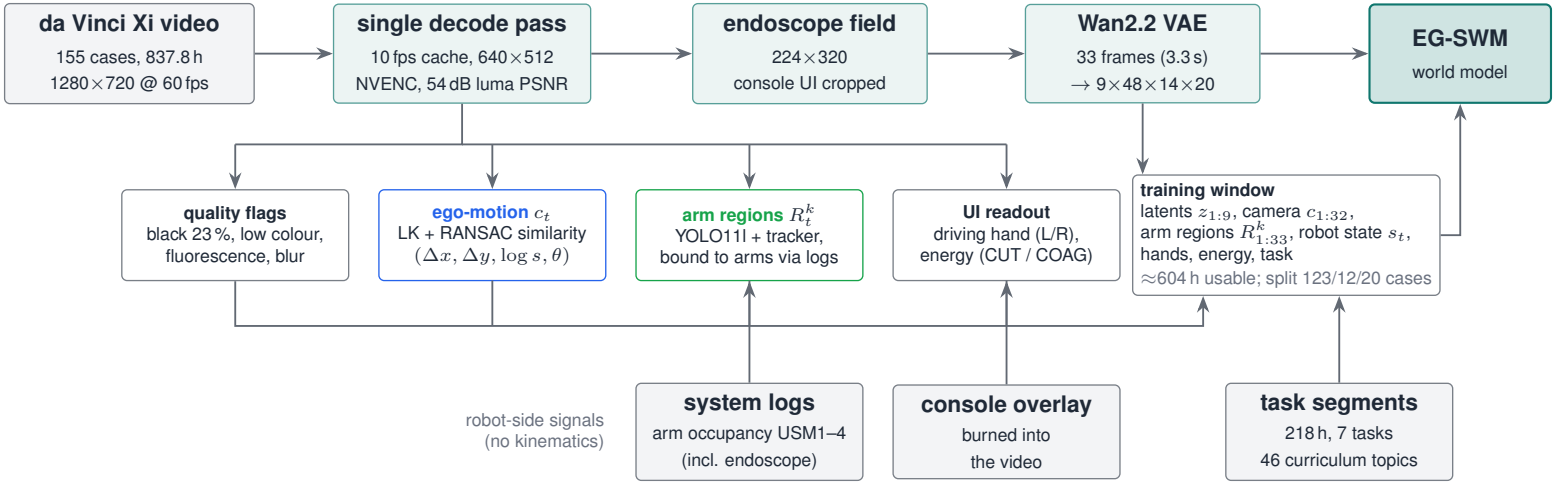

da Vinci Xi video
155 cases, 837.8 h
1280×720 @ 60 fps
single decode pass
10 fps cache, 640×512
NVENC, 54 dB luma PSNR
endoscope field
224×320
console UI cropped
Wan2.2 VAE
33 frames (3.3 s)
→ 9×48×14×20
EG-SWM
world model
quality flags
black 23 %, low colour,
fluorescence, blur
ego-motion $c_t$
LK + RANSAC similarity
$(\Delta x, \Delta y, \log s, \theta)$
arm regions $R_t^k$
YOLO11l + tracker,
bound to arms via logs
UI readout
driving hand (L/R),
energy (CUT / COAG)
training window
latents $z_{1:9}$, camera $c_{1:32}$,
arm regions $R_{1:33}^k$, robot state $s_t$,
hands, energy, task
≈604 h usable; split 123/12/20 cases
robot-side signals
(no kinematics)
system logs
arm occupancy USM1–4
(incl. endoscope)
console overlay
burned into
the video
task segments
218 h, 7 tasks
46 curriculum topics